LANDRESERVE

Product Requirements Document

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Document Type	Status	Target Market
Full PRD + Roadmap	**Active Development**	**Tanzania & East Africa**

1. Product Vision & Strategic Goals

LandReserve is the leading land marketplace for Tanzania and East Africa — an Airbnb-inspired platform where verified landowners list properties and trusted buyers or tenants transact with confidence. The platform removes the fraud, opacity, and friction that currently plague informal land markets in the region.

> 🎯 **North Star Metric**
> Number of completed (approved + paid) reservations per month. Everything in this document is prioritised to increase this number safely and sustainably.

1.1 Core Problems We Solve

Problem	How LandReserve Solves It
Fraud — fake listings, ghost sellers	KYC verification, government letter requirement, verified badge, 1-click report
No date transparency — is it available?	Real-time availability calendar, pending+approved blocks, conflict detection
Cash-only, no payment paper trail	M-Pesa STK push (roadmap), payment reference tracking, agreed-price lock
No trust between buyer and seller	Verified badge, review system, in-app messaging, escrow (roadmap)
Listing discovery is offline / word-of-mouth	Search + map, price filters, land-use categories, autocomplete
Agents charge opaque commissions	Direct owner-to-customer marketplace, transparent pricing

1.2 Technology Stack

Layer	Technology
Backend	Django 4.2+ (Python 3.10+)
Database	SQLite (dev) → PostgreSQL (prod)
Auth	Django Auth + django-allauth (Google OAuth)
Frontend Build	Bun + Tailwind CSS 3 + Bootstrap Icons
Maps	Leaflet.js + OpenStreetMap + MarkerCluster
SMS	Twilio API (graceful fallback if unconfigured)
Media Storage	Cloudinary (prod) / Local filesystem (dev)
Security	django-axes, django-ratelimit, django-csp 4.x, bleach
Deployment	Render.com (recommended) / any WSGI host

2. User Roles & Permissions

2.1 Role Matrix

Capability	Customer	Owner	Admin
Browse & search listings	✅	✅	✅
View land detail page	✅	✅	✅
Book / reserve land	✅	✅ (other owners)	✅
Cancel own reservation	✅	✅	✅
Wishlist	✅	✅	✅
Messaging	✅	✅	✅
Create / edit / delete listings	❌	✅	✅
Approve / reject bookings	❌	✅ (own listings)	✅
Submit KYC documents	❌	✅	N/A
Verify / suspend users	❌	❌	✅
Access admin portal	❌	❌	✅
Review KYC documents	❌	❌	✅
Switch modes (customer ↔ owner)	❌	✅	❌

3. Current Features (Implemented)

3.1 Land Listings

- Listing types: Rent, Sale
- Land use categories: Agricultural, Residential, Commercial, Industrial, Mixed Use
- Pricing: base price with Per Month / Per Year / Total (one-time) units
- Airbnb-style discounts: weekly (1+ week) and monthly (1+ month), 0–90%
- Minimum and maximum rental duration in days (rent listings only)
- GPS coordinates (lat/lng) for interactive map pin placement
- Cover photo + multi-image gallery (LandImage model with ordering)
- View count tracking — auto-incremented on every detail page visit
- Similar listings suggestion (by location or land use type)
- Active / inactive status flag (admin-editable)
- Location autocomplete API using Tanzania regional data + DB listings

3.2 Reservation / Booking System

- Full booking flow: date selection → validation → pending → approve/reject → SMS
- Overlap protection: pending AND approved reservations block date ranges
- Conflict check re-run at approval time (prevents race conditions)
- Anonymous booking: non-registered users can book by providing name/email/phone
- Booking status checker: anyone can look up status by email or phone
- Payment tracking: method, reference, amount paid, agreed price
- Payment methods: M-Pesa, Airtel Money, Tigo Pesa, Bank Transfer, Cash on Arrival
- Anti-spam: owners cannot book own listings; duplicate active booking prevented
- Approved booking cancellation notifies owner via SMS

3.3 Search & Discovery

- Filters: location, keyword, price range, listing type, land use, size range, availability, sort
- Sort options: newest, price low→high, price high→low, largest size
- Interactive Leaflet map with price marker popups and MarkerCluster
- Map syncs with active filters (availability filter applied to map pins)
- Pagination: 12 listings per page

3.4 Accounts & Authentication

- Standard registration with role selection (Customer / Land Owner)
- Google OAuth via django-allauth
- Brute-force protection: 5 failures → 1-hour lockout (django-axes)
- Rate limiting per endpoint (django-ratelimit)
- Suspended users blocked at login level (custom auth backend)
- Session-based auth with 1-week cookie expiry

3.5 KYC Verification

- Document upload: government ID or land title (JPG, PNG, WebP, PDF, max 10MB)
- "Barua ya Serikali za Mtaa" (government letter) with date validation — max 2 months old, no future dates
- Four-state flow: Not Submitted → Pending → Approved / Rejected
- Verified badge displayed on all listings of verified owners
- Unverified owners show cautionary grey badge

3.6 Messaging, Wishlist & Other

- Private threaded messaging between users, linkable to listings
- Unread message count badge in nav
- Wishlist: AJAX toggle, customer dashboard counter, dedicated list page
- Mode switching: owner-capable users switch between Customer and Owner UI
- Help center with categorised contact form and support email notification
- Listing report system (logged for admin review)
- Full admin portal: user management, KYC review, verification, suspension
- Django admin for database-level management
- /health/ endpoint for uptime monitoring
- SMS notifications on booking approval, rejection, and cancellation

4. Enhancement Roadmap — Brainstormed Ideas

> 💡 **How to read this section**
> Ideas are grouped by theme and tagged with priority: P1 = must-have for next milestone, P2 = high value / medium effort, P3 = nice-to-have / exploratory. Each idea includes the problem it solves, the proposed implementation approach, and estimated complexity.

4.1 Payment Gateway Integration

Currently payments are tracked manually (owner records a reference). Full integration will dramatically increase trust and conversion.

Feature	Implementation	Priority
M-Pesa STK Push (Daraja API)	Safaricom Daraja v2 API, store transaction ID on Reservation	P1
Airtel Money / Tigo Pesa	Respective mobile money APIs after M-Pesa validated	P2
Payment confirmation webhook	Receive async confirmation, auto-approve reservation on payment	P1
Deposit / partial payment	Deposit % field on Land model, balance due on approval	P2
Refund tracking	Refund reason + date fields, admin refund action in portal	P2
Platform commission (0–5%)	Commission rate in settings, deducted from agreed price, tracked on Reservation	P3
Stripe (card payments)	Stripe Checkout for international buyers; Stripe Connect for owner payouts	P3

> **M-Pesa STK Push Flow**
> 1. Customer initiates payment → 2. Backend calls Daraja STK Push API → 3. Customer receives prompt on phone and enters PIN → 4. Daraja sends webhook to /api/mpesa/callback/ → 5. Backend confirms transaction, sets payment_status=paid and auto-approves reservation → 6. SMS sent to customer and owner.

4.2 Calendar Availability View

Customers currently have no visual way to see which dates are taken before picking a range.

- Interactive date-range picker on the booking page showing blocked/available dates
- Monthly calendar view on land detail page (green = available, red = booked, grey = past)
- Blocked dates returned via a JSON API: GET /lands/<id>/availability/?month=2026-07
- Owner calendar view in dashboard — see all bookings per listing on a monthly grid
- iCal export (.ics) so owners can sync bookings to Google Calendar / Apple Calendar

- Minimum stay enforcement visually shown on the calendar (greys out short spans)
- [P1] Implement with Flatpickr or vanilla JS; backend API is a 1-day task

4.3 Reviews & Ratings System

Trust is the #1 blocker for first-time customers. Reviews give social proof.

- Post-stay review: customer can leave a review after a completed (approved + end_date passed) booking
- Star rating (1–5) + text comment (max 500 chars)
- Owner can reply to reviews (single reply only)
- Average rating and review count displayed on listing cards and detail pages
- Verified tag on reviews (confirms reviewer actually booked the land)
- [Data model] Review(land, customer, reservation, stars, text, owner_reply, created_at)
- [Admin] Ability to hide abusive reviews; report review flag
- [P1] Basic review submit + display; [P2] owner reply; [P3] helpful votes

4.4 Advanced Search & Discovery

The search bar is functional but misses several quality-of-life improvements that will increase engagement.

- [P1] Draw-on-map search — user draws a polygon on the map, lands inside it are shown in the panel
- [P1] Save search — logged-in users can save filter combinations with a name
- [P1] Saved search email alerts — notify user when a new listing matches their saved search
- [P2] Recent searches — last 5 search terms stored in localStorage, shown in autocomplete dropdown
- [P2] "Near me" button — uses browser geolocation API to search within a configurable radius
- [P2] Price histogram — bar chart above the price range slider showing distribution of listing prices
- [P2] Amenity/feature tags — filterable tags like "Water access", "Road frontage", "Electricity", "Fenced"
- [P3] AI-powered semantic search — "farmland near water in Morogoro" parsed by an LLM, mapped to filters
- [P3] Voice search — speech-to-text on mobile search bar

4.5 Listing Quality Improvements

Better listing content = higher conversion rate.

- [P1] Amenity checklist on listing form — "Water Access", "Road Frontage", "Electricity", "Fenced", "Irrigation", "Title Deed Available"
- [P1] Video upload support (max 50MB, MP4/MOV) stored on Cloudinary
- [P1] Land boundary drawing tool — Leaflet polygon draw plugin, stores GeoJSON in DB
- [P2] Listing completeness score — "Your listing is 70% complete. Add a video to reach more buyers." displayed to owner
- [P2] Auto-generated listing description suggestions using Claude AI API — owner approves/edits before saving
- [P2] Drone / satellite view toggle on the map — switch between OpenStreetMap and Esri satellite tiles
- [P2] Price comparison widget — "This land is 15% below average price in Arusha for agricultural land"
- [P3] Virtual tour / 360° photo support via Three.js or a Matterport embed
- [P3] Land value estimator — ML model trained on Tanzania land sale data

- [P3] Document storage — owner can attach PDF documents (survey plan, title deed copy) visible after booking approval

4.6 Owner Dashboard Analytics

Owners need actionable data to make decisions about pricing and listing management.

- [P1] Earnings chart — monthly bar chart of approved+paid reservation totals (Chart.js)
- [P1] Booking funnel — Views → Bookings → Approved → Paid conversion rates per listing
- [P1] Price benchmarking — "Your price is 20% above median in this area. Lower it to get more bookings."
- [P2] Response rate tracker — % of booking requests responded to within 24h (affects search ranking)
- [P2] Wishlist trend chart — how many users save this listing per week
- [P2] Top performing listings widget — ranked by views, bookings, earnings
- [P2] Export to CSV — owner downloads all bookings, payments, and customer contact info
- [P2] Export to PDF — printable booking confirmation / receipt for offline records
- [P3] Competitor pricing alerts — "3 new listings appeared in your area at lower prices"
- [P3] Revenue forecast — projected monthly earnings based on approved bookings

4.7 Notifications & Communication

The system currently sends SMS only. A full notification layer dramatically improves engagement.

- [P1] Email notifications — transactional email for booking requests, approval, rejection, cancellation using Django send_mail with HTML templates
- [P1] Email HTML templates — branded LandReserve email design matching the green colour system
- [P1] In-app notification bell — unread count badge, dropdown list of recent events (new booking, message, KYC status change)
- [P2] Push notifications — Web Push API for desktop; Firebase Cloud Messaging for mobile PWA
- [P2] Owner booking request summary — daily digest email "You have 3 pending booking requests"
- [P2] Customer booking reminder — 24h and 1h before booking start date via SMS + email
- [P2] WhatsApp notifications — Twilio WhatsApp API as alternative to SMS for Tanzania users
- [P3] Automated follow-up — after stay ends, auto-send "Please leave a review" email to customer

4.8 Mobile Experience & PWA

The majority of Tanzanian internet users are on mobile. A great mobile experience is mandatory.

- [P1] Progressive Web App (PWA) — add manifest.json and service worker so users can install from Chrome
- [P1] Offline mode — cache the last-viewed listings and booking status using Service Worker + Cache API
- [P1] Bottom navigation bar on mobile — replaces top nav for thumb-reachable UX on small screens
- [P2] Touch-optimised map — pinch-to-zoom, tap markers to open detail card instead of hover
- [P2] Mobile listing card swipe — left to wishlist, right to book (similar to Tinder-style UX for discovery)
- [P2] Image lazy loading with blur-up placeholder for slow connections
- [P2] Data-saver mode — strip gallery images on 2G/3G connections, show only primary image
- [P3] React Native / Flutter mobile app — dedicated iOS and Android apps

- [P3] SMS deep links — when owner receives a booking SMS, reply with Y to auto-approve via Twilio webhook

4.9 Trust, Safety & Anti-Fraud

Land fraud is the primary risk to platform reputation. Every feature here directly protects users.

- [P1] AI-powered listing moderation — flag listings with suspicious patterns (too cheap, generic photos, no GPS) using a simple classifier
- [P1] Duplicate listing detection — prevent same owner posting identical listing twice (title + location similarity check)
- [P1] Phone number verification — OTP SMS to customer phone before first booking goes through
- [P1] Image reverse search — detect stolen listing photos (Google Vision API or open-source perceptual hashing)
- [P2] Scam pattern database — internal blacklist of phone numbers, emails, and IP addresses from confirmed scam reports
- [P2] Two-factor authentication (2FA) — TOTP-based 2FA option in profile settings for owners and admins
- [P2] Escrow payment hold — platform holds M-Pesa payment until customer confirms land access received
- [P2] Dispute resolution workflow — customer files dispute, admin mediates, platform releases or refunds escrow
- [P3] Blockchain land title verification — integrate with Tanzania Land Registry API if/when available
- [P3] Insurance integration — optional booking insurance for high-value land transactions

4.10 Internationalisation & Localisation

- [P1] Kiswahili (Swahili) UI translation — full Django i18n with .po files for Swahili
- [P1] Currency display — show prices in TZS (Tanzanian Shilling) with correct formatting, not USD
- [P1] TZS / USD toggle — let users toggle between currencies with a live exchange rate
- [P2] Kenya support — KES currency, Safaricom M-Pesa Kenya integration, Kenya region list
- [P2] Uganda support — UGX, MTN Mobile Money
- [P2] Tanzania-specific land size units — add "Ekari" (local acre variant) and "Shamba" context labels
- [P3] Multi-language listing descriptions — owner can enter listing content in both English and Swahili

4.11 Land Inspection & Viewing Scheduling

- [P1] Physical inspection request — customer can request a site visit from the booking detail page
- [P1] Owner sets available inspection slots — calendar of available times; customer picks one
- [P2] In-app scheduling — integrated calendar with email/SMS reminders for both parties
- [P2] Video call inspection — Jitsi Meet embed for remote virtual inspections (useful for diaspora buyers)
- [P3] Licensed surveyor booking — connect customer to a verified surveyor from a partner network

4.12 Land Comparison Tool

- [P2] Side-by-side comparison — user selects up to 3 listings, sees price/size/location/features in a table

- [P2] Comparison link — sharable URL that opens the same 3 listings in comparison view
- [P2] Price per acre/hectare normalisation — calculated and shown in comparison for fair evaluation
- [P3] Print comparison PDF — exportable PDF report for taking to a bank or legal advisor

4.13 Admin Power Tools

- [P1] System-level statistics dashboard — total users, listings, reservations, revenue this month vs last month
- [P1] Bulk KYC review — table with approve/reject toggles for multiple pending documents at once
- [P1] Audit log — every admin action (verify, suspend, delete) logged with timestamp and actor
- [P2] Listing moderation queue — flagged/reported listings appear in a single review queue with approve/remove actions
- [P2] Revenue dashboard — platform commission totals, payout status, monthly trends
- [P2] User cohort analytics — registration trend, activation rate, churn by user role
- [P2] Broadcast message — admin can send a notification to all users or a filtered segment
- [P3] A/B test framework — flag percentage of users to see experimental features

4.14 Public REST API & Integrations

- [P2] REST API — Django REST Framework with JWT auth for /api/v1/lands/, /api/v1/reservations/ endpoints
- [P2] API documentation — auto-generated Swagger/OpenAPI docs at /api/docs/
- [P2] Webhook system — owners can register a webhook URL to receive booking events in their own systems
- [P2] WhatsApp Business integration — send booking details and status updates via WhatsApp Business API
- [P3] Google Maps Places API — richer location autocomplete with satellite thumbnails
- [P3] Tanzania Land Registry API integration — verify legal land title numbers in real time
- [P3] Partner agent portal — licensed real estate agents get a sub-portal with their own branded listing pages

5. Data Models (Current + Planned)

5.1 Current Models

Model	Key Fields
accounts.User	username, email, role, is_owner, is_verified, is_suspended, phone, bio, profile_picture, kyc_document, kyc_status, kyc_notes, govt_letter, govt_letter_date
lands.Land	owner, title, description, location, latitude, longitude, listing_type, size, size_unit, land_use, price, price_unit, weekly_discount, monthly_discount, min/max_duration_days, contact_phone/email, image, is_active, view_count
lands.LandImage	land (FK), image, caption, is_primary, order, created_at
lands.Reservation	land, customer (FK, nullable), customer_name/email/phone, start_date, end_date, status, payment_status, payment_method, payment_reference, amount_paid, agreed_price, notes, booking_date
lands.Wishlist	user (FK), land (FK), created_at — unique_together (user, land)
lands.Message	sender, recipient, land (FK nullable), subject, body, is_read, created_at

5.2 Planned New Models

Model	Key Fields
lands.Review	land (FK), customer (FK), reservation (FK), stars (1–5), text, owner_reply, is_hidden, created_at
lands.LandAmenity	land (FK), amenity_type (water/road/electricity/fence/irrigation/title_deed), added_at
lands.LandBoundary	land (OneToOne), geojson (TextField storing GeoJSON polygon), created_at
lands.InspectionSlot	land (FK), owner, datetime, duration_minutes, is_booked
lands.InspectionBooking	slot (FK), customer (FK), reservation (FK nullable), notes, status (pending/confirmed/completed)
lands.Notification	recipient (FK), type, title, body, action_url, is_read, created_at
lands.SavedSearch	user (FK), name, filters_json (JSONField), alert_enabled, last_notified_at
payments.Payment	reservation (FK), method, amount, currency, transaction_id, status, initiated_at, confirmed_at, raw_response (JSONField)
payments.Commission	payment (FK), rate, amount, paid_out, payout_reference

6. URL / API Reference (Current)

6.1 Public

URL	Description
/	Landing / browse listings (land_list view)
/lands/search/	Search with filters
/lands/<id>/	Land detail page
/lands/<id>/book/	Booking form
/lands/check-status/	Anonymous booking lookup by email/phone
/lands/help/	Help centre contact form
/health/	Health check endpoint (JSON)
/accounts/google/login/	Google OAuth initiation (allauth)

6.2 Authenticated (Customer + Owner)

URL	Description
/lands/dashboard/customer/	Customer dashboard
/lands/reservations/	My bookings
/lands/wishlist/	My wishlist
/lands/<id>/wishlist/	Toggle wishlist (POST, AJAX-friendly)
/lands/messages/	Message inbox
/lands/messages/<user_id>/	Message thread
/accounts/profile/edit/	Edit profile, upload KYC
/lands/switch-mode/	Toggle customer ↔ owner mode (POST)

6.3 Owner-Only

URL	Description
/lands/dashboard/	Owner dashboard + analytics
/lands/add/	Create new listing
/lands/<id>/edit/	Edit listing
/lands/<id>/delete/	Delete listing
/lands/reservations/manage/	All booking management
/lands/reservations/<id>/status/<action>/	Approve / reject / cancel booking (POST)
/lands/reservations/<id>/payment/	Record payment (POST)

7. Security Requirements

7.1 Current Security Layer

Control	Implementation
Input sanitisation	bleach.clean() + strip_tags() on all user text inputs
SQL injection	Django ORM (parameterised queries — no raw SQL)
XSS	Django auto-escaping + CSP headers via django-csp 4.x
CSRF	Django CsrfViewMiddleware on all POST forms
Brute force	django-axes: 5 failures → 1h lockout per IP
Rate limiting	django-ratelimit per endpoint (5–30 req/min)
File upload	content_type whitelist + 5MB max size enforced in forms
Clickjacking	X-Frame-Options: DENY
Transport security	HTTPS redirect + HSTS (1 year) in production
Session security	HttpOnly + SameSite=Lax cookies; 1-week expiry
Referrer policy	strict-origin-when-cross-origin

7.2 Planned Security Improvements

- [P1] Phone number OTP verification before first booking — prevent throwaway bookings
- [P1] Two-factor authentication (TOTP) option for owners and admins
- [P1] Audit log table — store every sensitive action with actor, timestamp, affected object
- [P2] AI listing fraud detection — flag listings matching scam patterns automatically
- [P2] Escrow payment hold — funds held until customer confirms access
- [P2] GDPR/POPIA compliance module — data export, account deletion, consent tracking
- [P3] Penetration testing — scheduled quarterly pen test against production
- [P3] Bug bounty programme — public responsible disclosure policy

8. Deployment & Infrastructure

8.1 Recommended Stack

Layer	Recommended	Alternative
App hosting	Render.com Web Service	Railway, Heroku, Fly.io
Database	Render PostgreSQL (free tier)	Supabase, Neon.tech
Media storage	Cloudinary (free 25GB)	AWS S3 + CloudFront
Email	Resend.com or SendGrid	AWS SES, Mailgun
SMS	Twilio	Africa's Talking (cheaper in Africa)
Monitoring	Sentry (free tier)	Rollbar, Datadog
Uptime	UptimeRobot (pings /health/)	Better Uptime, Freshping

8.2 Build & Start Commands

Build command (Render):

pip install -r requirements.txt && bun install && bun run build && python manage.py collectstatic --noinput && python manage.py migrate

Start command:

gunicorn land_reservation.wsgi:application --workers 2 --timeout 120

8.3 Required Environment Variables

Variable	Description
SECRET_KEY	Random 50-char Django secret — generate with: python -c "from django.core.management.utils import get_random_secret_key; print(get_random_secret_key())"
DEBUG	Set to False in production
ALLOWED_HOSTS	Comma-separated list: yourapp.onrender.com
DATABASE_URL	PostgreSQL connection string from Render
SUPPORT_EMAIL	Email address for receiving help centre submissions
TWILIO_ACCOUNT_SID	Twilio SID for SMS notifications
TWILIO_AUTH_TOKEN	Twilio auth token
TWILIO_PHONE_NUMBER	Twilio sender number (+E.164 format)
GOOGLE_CLIENT_ID	Google OAuth client ID for social login
GOOGLE_CLIENT_SECRET	Google OAuth client secret
CLOUDINARY_CLOUD_NAME	Cloudinary cloud name for media storage
CLOUDINARY_API_KEY	Cloudinary API key

Variable	Description
CLOUDINARY_API_SECRET	Cloudinary API secret

9. Implementation Priority Roadmap

How priorities are set
P1 = directly increases completed reservations (North Star); P2 = increases trust and retention; P3 = expansion and scale features. Do P1 items before touching P2.

Phase 1 — Foundation (Next 30 Days)

Feature	Section	Effort
M-Pesa STK Push payment integration	4.1	L (1–2 weeks)
Calendar availability view on booking page	4.2	M (3–5 days)
Email transactional notifications (booking events)	4.7	M (2–3 days)
Review & rating system (submit + display)	4.3	M (4–6 days)
Amenity checklist on listing form	4.5	S (1–2 days)
Phone OTP verification before first booking	7.2	M (2–3 days)
Swahili translation (i18n)	4.10	L (1 week)
TZS currency display (not USD)	4.10	S (1 day)
In-app notification bell + Notification model	4.7	M (3–4 days)

Phase 2 — Growth (Days 31–90)

Feature	Section	Effort
Owner dashboard earnings chart	4.6	S (2 days)
Saved search + email alerts	4.4	M (4–5 days)
PWA (manifest.json + service worker)	4.8	M (2–3 days)
Land boundary drawing (Leaflet Draw plugin)	4.5	M (3 days)
Video upload support (Cloudinary)	4.5	M (2 days)
Price comparison widget on listing detail	4.5	S (1 day)
Inspection slot scheduling	4.11	M (4–5 days)
2FA (TOTP) for owners & admins	7.2	M (2–3 days)
Bulk KYC review in admin portal	4.13	S (1–2 days)
Booking export to CSV/PDF	4.6	S (2 days)
WhatsApp notifications (Twilio)	4.7	M (2 days)
Owner response rate tracker	4.6	S (1 day)

Phase 3 — Scale (Days 91+)

- REST API (Django REST Framework) + Swagger docs
- Land comparison tool (side-by-side 3 listings)
- AI-powered listing fraud detection
- Escrow payment hold + dispute resolution workflow
- Kenya & Uganda market expansion (currency, mobile money, regions)
- React Native / Flutter mobile app
- AI listing description generator (Claude API)
- Platform commission system + owner payout tracking
- Tanzania Land Registry API integration
- Quarterly penetration testing

10. Glossary

Term	Definition
KYC	Know Your Customer — identity verification process for land owners
Barua ya Serikali za Mtaa	Letter from local government authority — required for owner verification
M-Pesa / Daraja	Safaricom mobile money service; Daraja is the API for STK Push payments
STK Push	SIM Toolkit Push — triggers a payment prompt on the customer's phone
Rate Limiting	Restricting the number of requests a single IP/user can make per time period
Brute Force	Repeated login attempts to guess a password; blocked by django-axes
PWA	Progressive Web App — web app installable on mobile like a native app
CSP	Content Security Policy — browser security header preventing XSS attacks
CSRF	Cross-Site Request Forgery — attack prevented by Django's CsrfViewMiddleware
Escrow	Platform holds payment until both parties complete the transaction
TOTP	Time-based One-Time Password — used for 2-factor authentication (Google Authenticator)
GeoJSON	JSON format for encoding geographic data; used for land boundary polygons
North Star Metric	The single metric that best captures the platform's core value delivery
P1 / P2 / P3	Priority levels: P1=must-have, P2=high-value, P3=nice-to-have

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Built for Tanzania & East Africa — the land marketplace that communities trust